

Code ▼

# Keio - Esophageal Neo-FLOT\_Clinical Final analysis 04092026

```
library(swimplot) library(coxphf) library(grid) library(gtable) library(readr) library(mosaic) library(dplyr) library(survival)
library(survminer) library(ggplot2) library(scales) library(coxphf) library(ggthemes) library(tidyverse)
library(gtsummary) library(flextable) library(parameters) library(car) library(ComplexHeatmap) library(tidyverse)
library(readxl) library(survival) library(janitor) library(openxlsx) library(writexl) library(rms) library(pROC) library(DT)
```

#ctDNA Detection rate by Stage and Window

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```
#Baseline
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Keio Neo FLOT_Clinical Data.csv")
circ_data$ctDNA.Base <- factor(circ_data$ctDNA.Base, levels=c("NEGATIVE","POSITIVE"))
circ_data$Stage <- factor(circ_data$Stage, levels=c("I","II","III","IV"))
circ_data <- subset(circ_data, ctDNA.Base %in% c("NEGATIVE", "POSITIVE"))
positive_counts_by_stage <- aggregate(circ_data$ctDNA.Base == "POSITIVE", by=list(circ_data$Stage), FUN=sum)
total_counts_by_stage <- aggregate(circ_data$ctDNA.Base, by=list(circ_data$Stage), FUN=length)
combined_data <- data.frame(
  Stage = total_counts_by_stage$Group.1,
  Total_Count = total_counts_by_stage$x,
  Positive_Count = positive_counts_by_stage$x,
  Rate = (positive_counts_by_stage$x / total_counts_by_stage$x) * 100 # Convert to percentage
)
combined_data$Rate <- sprintf("%.2f%%", combined_data$Rate)
overall_total_count <- nrow(circ_data)
overall_positive_count <- nrow(circ_data[circ_data$ctDNA.Base == "POSITIVE",])
overall_positivity_rate <- (overall_positive_count / overall_total_count) * 100 # Convert to percentage
overall_row <- data.frame(
  Stage = "Overall",
  Total_Count = overall_total_count,
  Positive_Count = overall_positive_count,
  Rate = sprintf("%.2f%%", overall_positivity_rate)
)
combined_data <- rbind(combined_data, overall_row)
print(combined_data)
```

|   | Stage   | Total_Count | Positive_Count | Rate    |
|---|---------|-------------|----------------|---------|
| 1 | I       | 3           | 2              | 66.67%  |
| 2 | II      | 7           | 7              | 100.00% |
| 3 | III     | 19          | 19             | 100.00% |
| 4 | IV      | 5           | 5              | 100.00% |
| 5 | Overall | 34          | 33             | 97.06%  |

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```
#post-NAC Window
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Keio Neo FLOT_Clinical Data.csv")
circ_data$ctDNA.Base <- factor(circ_data$ctDNA.postNAC, levels=c("NEGATIVE","POSITIVE"))
circ_data$Stage <- factor(circ_data$Stage, levels=c("I","II","III","IV"))
circ_data <- subset(circ_data, ctDNA.postNAC %in% c("NEGATIVE", "POSITIVE"))
positive_counts_by_stage <- aggregate(circ_data$ctDNA.postNAC == "POSITIVE", by=list(circ_data$Stage), FUN=sum)
total_counts_by_stage <- aggregate(circ_data$ctDNA.postNAC, by=list(circ_data$Stage), FUN=length)
combined_data <- data.frame(
  Stage = total_counts_by_stage$Group.1,
  Total_Count = total_counts_by_stage$x,
  Positive_Count = positive_counts_by_stage$x,
  Rate = (positive_counts_by_stage$x / total_counts_by_stage$x) * 100 # Convert to percentage
)
combined_data$Rate <- sprintf("%.2f%%", combined_data$Rate)
overall_total_count <- nrow(circ_data)
overall_positive_count <- nrow(circ_data[circ_data$ctDNA.postNAC == "POSITIVE",])
overall_positivity_rate <- (overall_positive_count / overall_total_count) * 100 # Convert to percentage
overall_row <- data.frame(
  Stage = "Overall",
  Total_Count = overall_total_count,
  Positive_Count = overall_positive_count,
  Rate = sprintf("%.2f%%", overall_positivity_rate)
)
combined_data <- rbind(combined_data, overall_row)
print(combined_data)
```

|   | Stage   | Total_Count | Positive_Count | Rate   |
|---|---------|-------------|----------------|--------|
| 1 | I       | 3           | 1              | 33.33% |
| 2 | II      | 7           | 2              | 28.57% |
| 3 | III     | 18          | 7              | 38.89% |
| 4 | IV      | 5           | 4              | 80.00% |
| 5 | Overall | 33          | 14             | 42.42% |

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```

#MRD Window
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Keio Neo FLOT_Clinical Data.csv")
circ_data$ctDNA.MRD <- factor(circ_data$ctDNA.MRD, levels=c("NEGATIVE","POSITIVE"))
circ_data$Stage <- factor(circ_data$Stage, levels=c("I","II","III"))
circ_data <- subset(circ_data, ctDNA.MRD %in% c("NEGATIVE", "POSITIVE"))
positive_counts_by_stage <- aggregate(circ_data$ctDNA.MRD == "POSITIVE", by=list(circ_data$Stage), FUN=sum)
total_counts_by_stage <- aggregate(circ_data$ctDNA.MRD, by=list(circ_data$Stage), FUN=length)
combined_data <- data.frame(
  Stage = total_counts_by_stage$Group.1,
  Total_Count = total_counts_by_stage$x,
  Positive_Count = positive_counts_by_stage$x,
  Rate = (positive_counts_by_stage$x / total_counts_by_stage$x) * 100 # Convert to percentage
)
combined_data$Rate <- sprintf("%.2f%%", combined_data$Rate)
overall_total_count <- nrow(circ_data)
overall_positive_count <- nrow(circ_data[circ_data$ctDNA.MRD == "POSITIVE",])
overall_positivity_rate <- (overall_positive_count / overall_total_count) * 100 # Convert to percentage
overall_row <- data.frame(
  Stage = "Overall",
  Total_Count = overall_total_count,
  Positive_Count = overall_positive_count,
  Rate = sprintf("%.2f%%", overall_positivity_rate)
)
combined_data <- rbind(combined_data, overall_row)
print(combined_data)

```

|   | Stage   | Total_Count | Positive_Count | Rate   |
|---|---------|-------------|----------------|--------|
| 1 | I       | 3           | 0              | 0.00%  |
| 2 | II      | 6           | 1              | 16.67% |
| 3 | III     | 16          | 7              | 43.75% |
| 4 | Overall | 29          | 9              | 31.03% |

#Demographics Table

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```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Keio Neo FLOT_Clinical Data.csv")

circ_data_subset <- circ_data %>%
  select(
    Age,
    Sex,
    ECOG,
    Location,
    Stage,
    Resection,
    pGrade,
    Recurrence,
    OS.Event,
    OS.months) %>%
  mutate(
    Age = as.numeric(Age),
    Sex = factor(Sex, levels = c("Male", "Female")),
    ECOG = factor(ECOG, levels = c("0", "1")),
    Location = factor(Location, levels = c("Lt", "Mt", "Ut")),
    Stage = factor(Stage, levels = c("I","II","III","IV")),
    Resection = factor(Resection, levels = c("R0","R1")),
    pGrade = factor(pGrade, levels = c("Grade0","Grade1","Grade2","Grade3")),
    Recurrence = factor(Recurrence, levels = c("TRUE","FALSE")),
    OS.Event = factor(OS.Event, levels = c("TRUE","FALSE")),
    OS.months = as.numeric(OS.months))
table1 <- circ_data_subset %>%
  tbl_summary(
    statistic = list(
      all_continuous() ~ "{median} ({min} - {max})",
      all_categorical() ~ "{n} ({p}%)" ) %>%
    bold_labels()
  )
table1
```

| Characteristic                         | N = 34 <sup>1</sup> |
|----------------------------------------|---------------------|
| <b>Age</b>                             | 66 (44 - 80)        |
| <b>Sex</b>                             |                     |
| Male                                   | 27 (79%)            |
| Female                                 | 7 (21%)             |
| <b>ECOG</b>                            |                     |
| 0                                      | 32 (94%)            |
| 1                                      | 2 (5.9%)            |
| <sup>1</sup> Median (Min - Max); n (%) |                     |

| Characteristic                         | N = 34 <sup>1</sup> |
|----------------------------------------|---------------------|
| <b>Location</b>                        |                     |
| Lt                                     | 14 (41%)            |
| Mt                                     | 18 (53%)            |
| Ut                                     | 2 (5.9%)            |
| <b>Stage</b>                           |                     |
| I                                      | 3 (8.8%)            |
| II                                     | 7 (21%)             |
| III                                    | 19 (56%)            |
| IV                                     | 5 (15%)             |
| <b>Resection</b>                       |                     |
| R0                                     | 29 (94%)            |
| R1                                     | 2 (6.5%)            |
| Unknown                                | 3                   |
| <b>pGrade</b>                          |                     |
| Grade0                                 | 1 (3.2%)            |
| Grade1                                 | 15 (48%)            |
| Grade2                                 | 12 (39%)            |
| Grade3                                 | 3 (9.7%)            |
| Unknown                                | 3                   |
| <b>Recurrence</b>                      |                     |
| TRUE                                   | 18 (53%)            |
| FALSE                                  | 16 (47%)            |
| <b>OS.Event</b>                        |                     |
| TRUE                                   | 12 (35%)            |
| FALSE                                  | 22 (65%)            |
| <b>OS.months</b>                       | 26 (8 - 52)         |
| <sup>1</sup> Median (Min - Max); n (%) |                     |

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```
fit1 <- as_flex_table(  
  table1,  
  include = everything(),  
  return_calls = FALSE,  
  strip_md_bold = TRUE)  
fit1
```

| Characteristic   | N = 34 <sup>1</sup> |
|------------------|---------------------|
| <b>Age</b>       | 66 (44 - 80)        |
| <b>Sex</b>       |                     |
| Male             | 27 (79%)            |
| Female           | 7 (21%)             |
| <b>ECOG</b>      |                     |
| 0                | 32 (94%)            |
| 1                | 2 (5.9%)            |
| <b>Location</b>  |                     |
| Lt               | 14 (41%)            |
| Mt               | 18 (53%)            |
| Ut               | 2 (5.9%)            |
| <b>Stage</b>     |                     |
| I                | 3 (8.8%)            |
| II               | 7 (21%)             |
| III              | 19 (56%)            |
| IV               | 5 (15%)             |
| <b>Resection</b> |                     |
| R0               | 29 (94%)            |
| R1               | 2 (6.5%)            |
| Unknown          | 3                   |
| <b>pGrade</b>    |                     |
| Grade0           | 1 (3.2%)            |
| Grade1           | 15 (48%)            |
| Grade2           | 12 (39%)            |
| Grade3           | 3 (9.7%)            |

<sup>1</sup>Median (Min - Max); n (%)

| Characteristic | N = 34 <sup>1</sup> |
|----------------|---------------------|
| Unknown        | 3                   |
| Recurrence     |                     |
| TRUE           | 18 (53%)            |
| FALSE          | 16 (47%)            |
| OS.Event       |                     |
| TRUE           | 12 (35%)            |
| FALSE          | 22 (65%)            |
| OS.months      | 26 (8 - 52)         |

<sup>1</sup>Median (Min - Max); n (%)

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```
save_as_docx(fit1, path= "~/Downloads/1. Keio ESCC_Demographics Table.docx")
```

#Heatmap with Clinical & Genomics Factors

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```

rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Keio Neo FLOT_Clinical Data.csv")
circ_data <- circ_data %>% arrange(Stage)
circ_datadf <- as.data.frame(circ_data)

ha <- HeatmapAnnotation(
  Stage = circ_data$Stage,
  Sex = circ_data$Sex,
  Location = circ_data$Location,
  Grade = circ_data$Grade,
  ctDNA.Base = circ_data$ctDNA.Base,
  ctDNA.postNAC = circ_data$ctDNA.postNAC,
  ctDNA.MRD = circ_data$ctDNA.MRD,
  ctDNA.surveillance = circ_data$ctDNA.surveillance,
  EFS.Event = circ_data$EFS.Event,
  OS.Event = circ_data$OS.Event,

  col = list(Stage = c("I" = "seagreen1", "II" = "seagreen", "III" = "orange", "IV" = "purple"),
    Sex = c("Female" = "goldenrod", "Male" = "blue4"),
    Location = c("Lt" = "brown", "Mt" = "darkgreen", "Ut" = "orange4"),
    Grade = c("Grade1b/2/3" = "cornflowerblue", "Grade0/1a" = "darkmagenta"),
    ctDNA.Base = c("POSITIVE" = "red3", "NEGATIVE" = "blue"),
    ctDNA.postNAC = c("POSITIVE" = "red3", "NEGATIVE" = "blue"),
    ctDNA.MRD = c("POSITIVE" = "red3", "NEGATIVE" = "blue"),
    EFS.Event = c("TRUE" = "red3", "FALSE" = "blue"),
    OS.Event = c("TRUE" = "black", "FALSE" = "gray")
  ))
ht <- Heatmap(matrix(nrow = 0, ncol = length(circ_data$Stage)), show_row_names = FALSE,
  cluster_rows = F, cluster_columns = FALSE, top_annotation = ha)
pdf("heatmap.pdf", width = 7, height = 7)
draw(ht, annotation_legend_side = "bottom")
dev.off()

```

#Overview Plot

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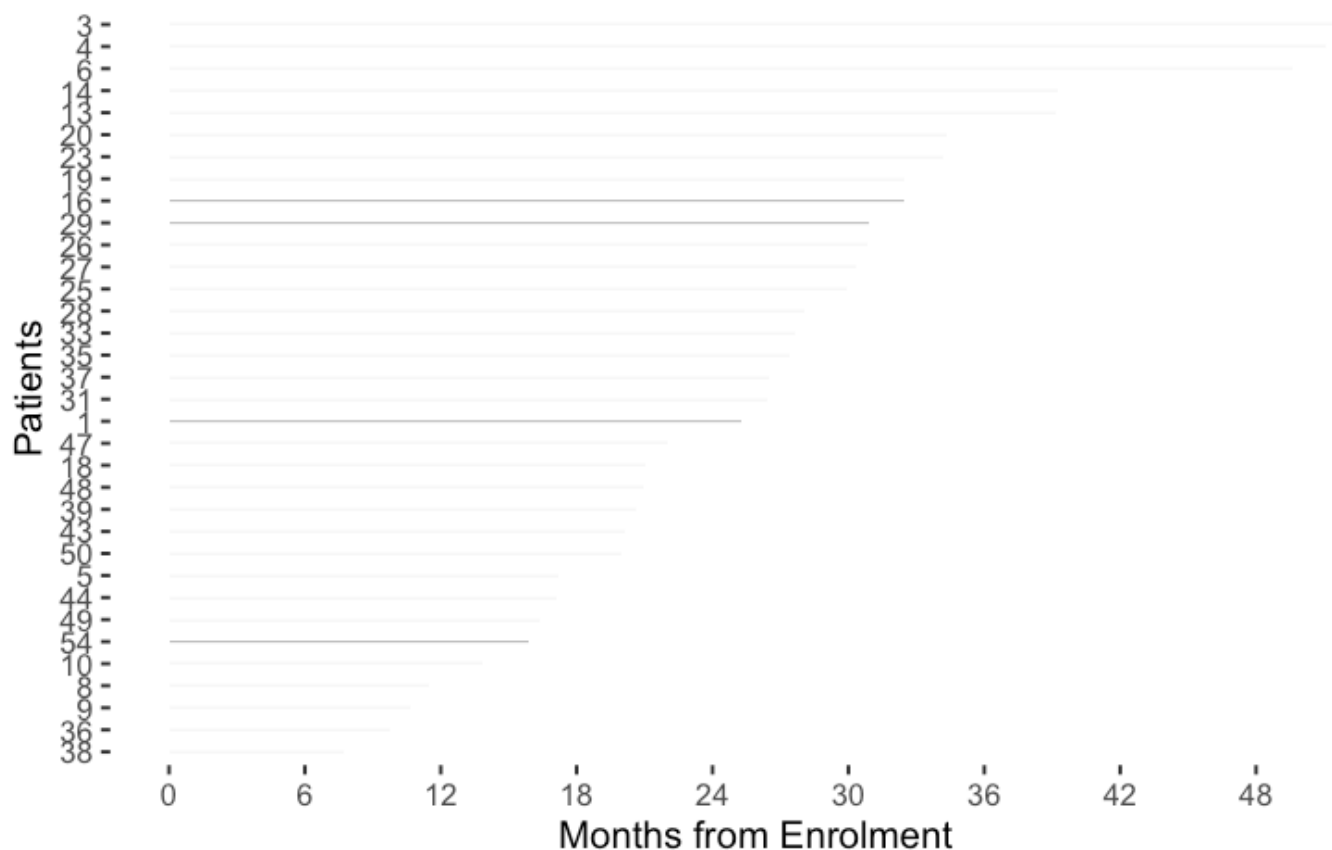


```

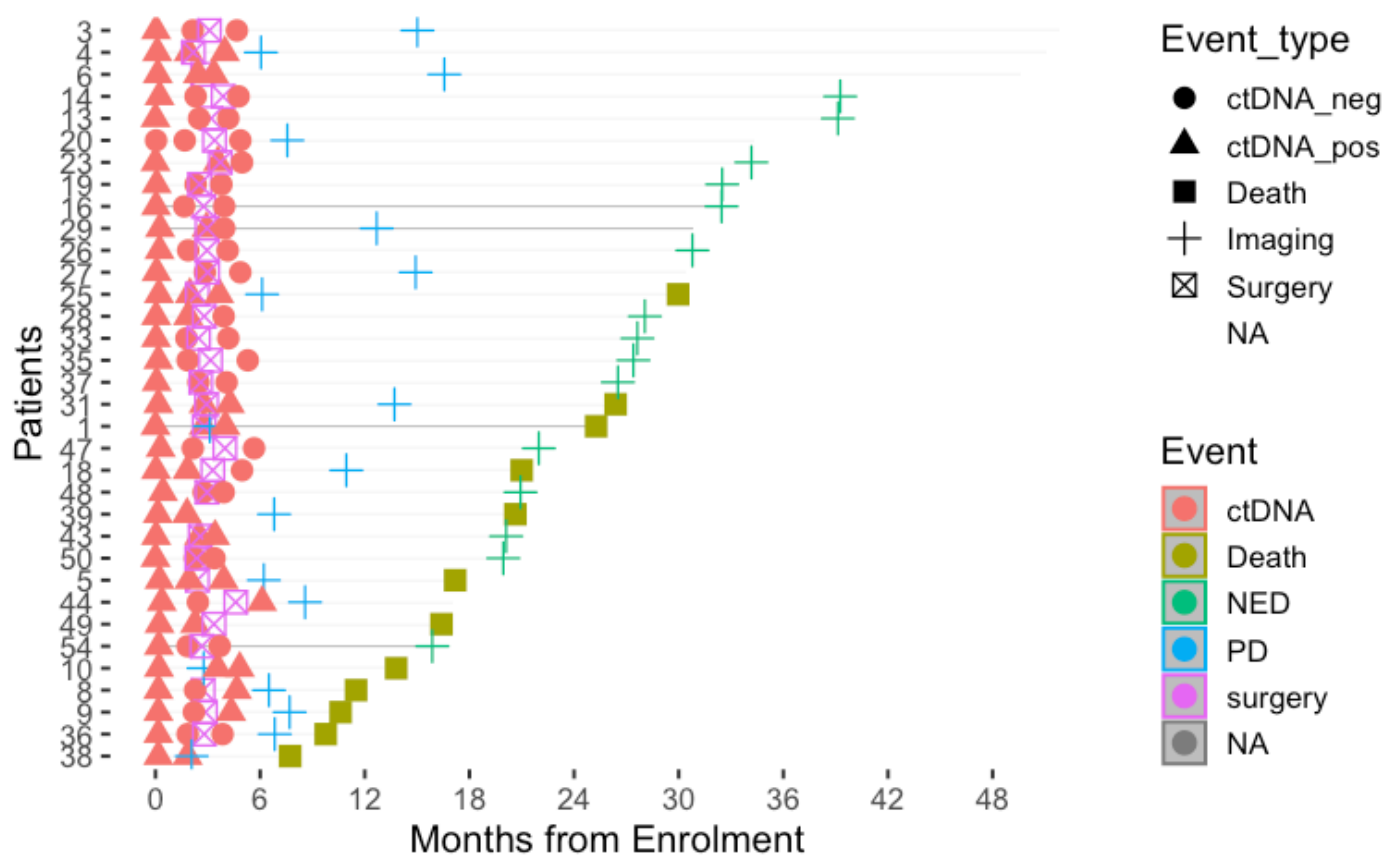
setwd("~/Downloads")
clinstage<- read.csv("Keio Neo FLOT_OP.csv")
clinstage_df<- as.data.frame(clinstage)

#Display the swimmer plot with the label box
oplot<-swimmer_plot(df=clinstage_df,
                    id='PatientName',
                    end='fu.diff.months',
                    fill='gray',
                    width=.01,
                    base_size = 14)
oplot <- oplot + theme(panel.border = element_blank())
oplot <- oplot + scale_y_continuous(breaks = seq(-12, 48, by = 6))
oplot <- oplot + labs(x ="Patients" , y="Months from Enrolment")
oplot

```


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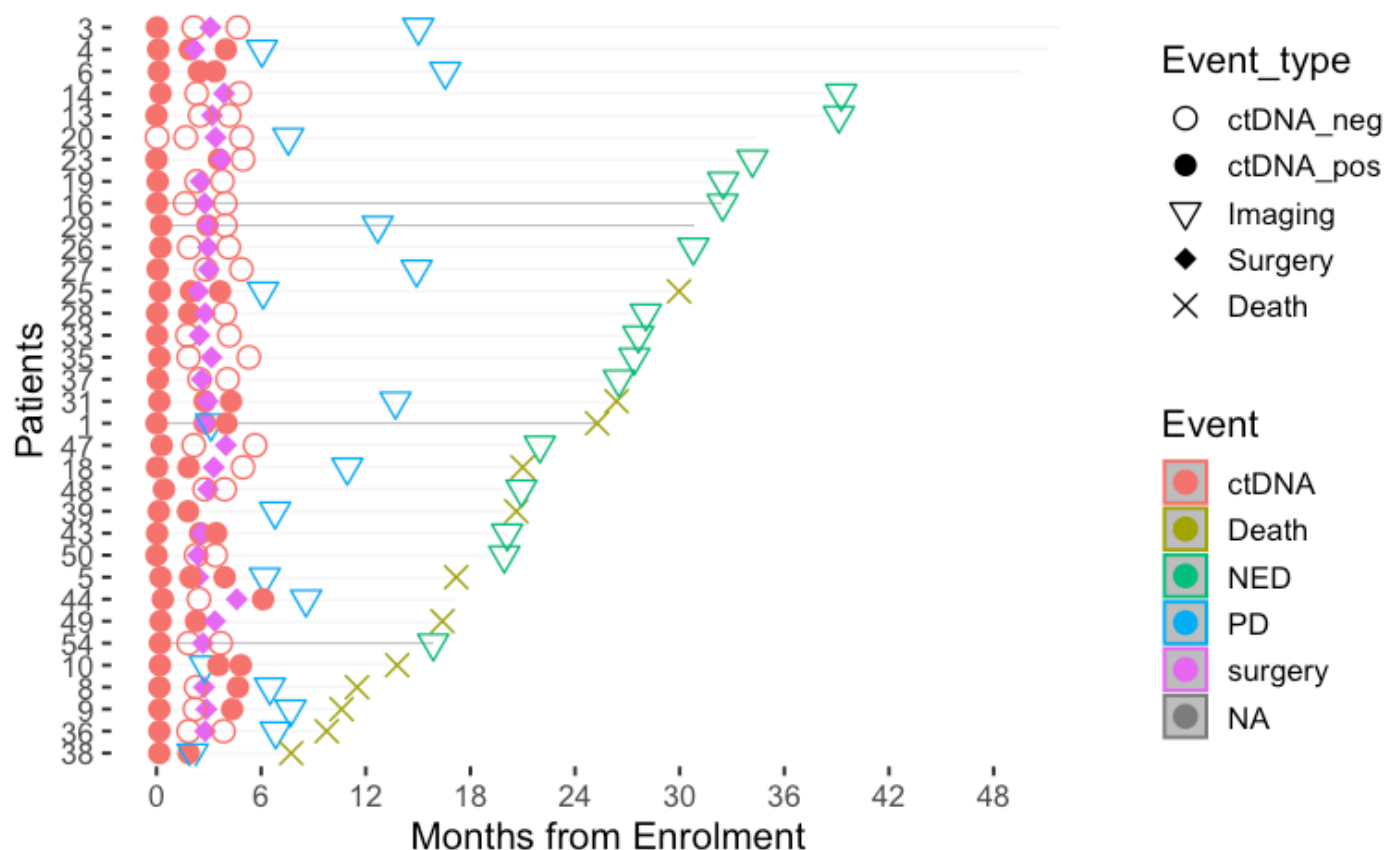
```
##plot events
oplot_ev1 <- oplot + swimmer_points(df_points=clinstage_df,
                                   id='PatientName',
                                   time='date.diff.months',
                                   name_shape = 'Event_type',
                                   name_col = 'Event',
                                   size=3.5,fill='black',
                                   #col='darkgreen'
)
oplot_ev1
```


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```
#Shape customization to Event_type

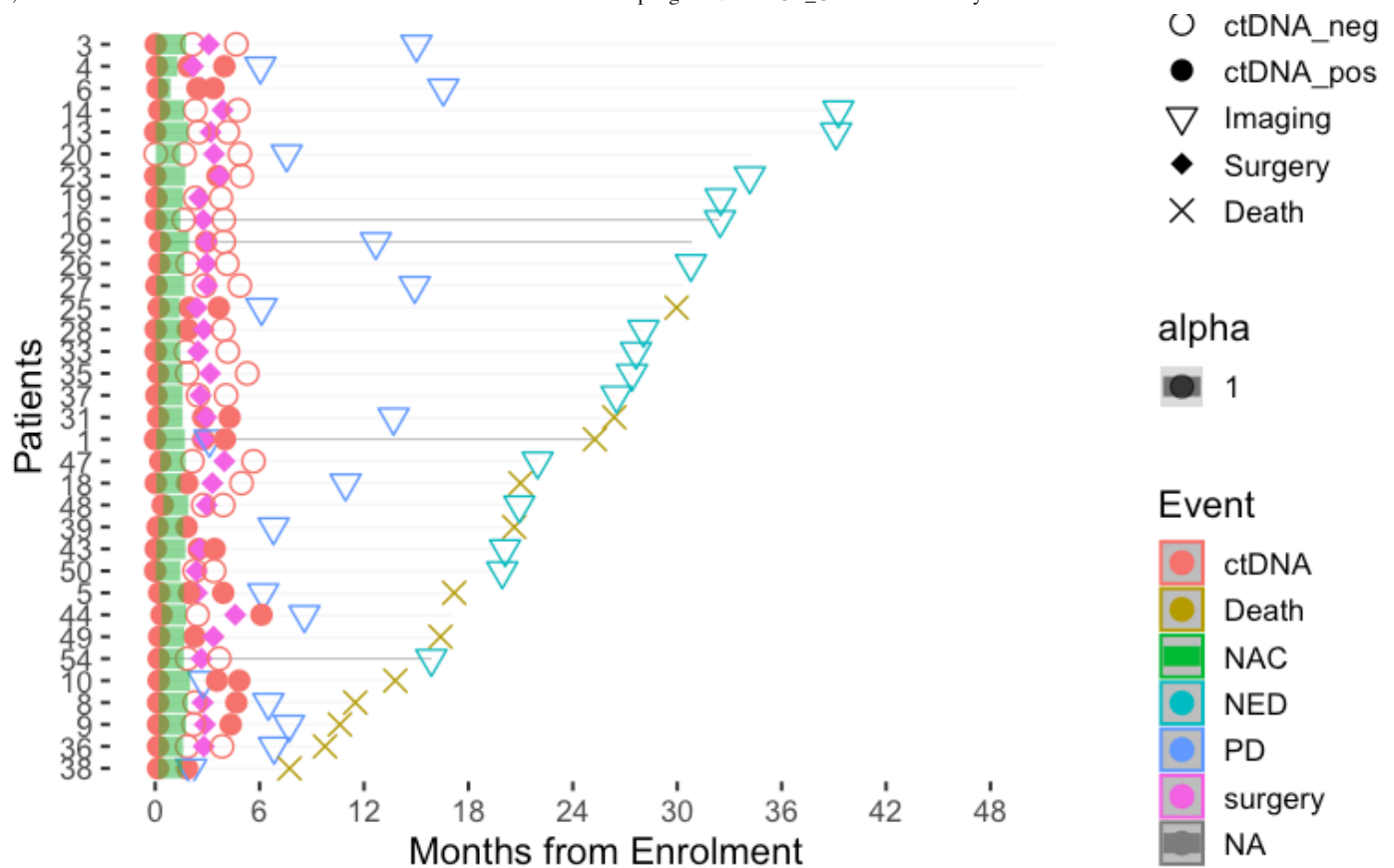
oplot_ev1.1 <- oplot_ev1 + ggplot2::scale_shape_manual(name="Event_type",values=c(1,16,
6,18,4),breaks=c('ctDNA_neg','ctDNA_pos', 'Imaging','Surgery', 'Death'))

oplot_ev1.1
```

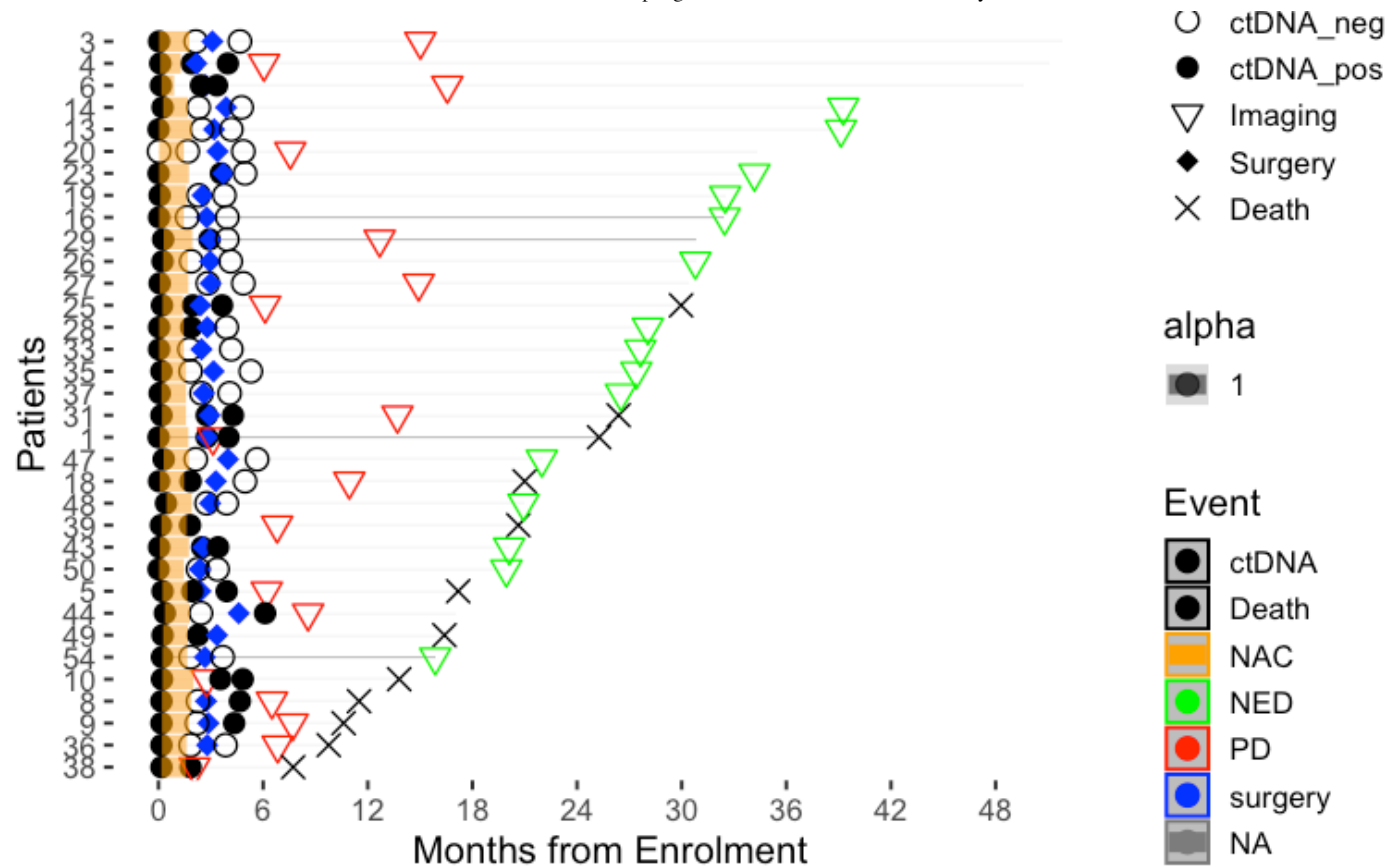

[Hide](#)

```
#plot treatment
```

```
oplot_ev2 <- oplot_ev1.1 + swimmer_lines(df_lines=clinstage_df,
                                         id='PatientName',
                                         start='Tx_start.months',
                                         end='Tx_end.months',
                                         name_col='Tx_type',
                                         size=3.5,
                                         name_alpha = 1.0)
oplot_ev2 <- oplot_ev2 + guides(linetype = guide_legend(override.aes = list(size = 5, color = "black")))
oplot_ev2
```


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```
#colour customization
oplot_ev2.2 <- oplot_ev2 + ggplot2::scale_color_manual(name="Event", values=c("black", "black", "orange", "green", "red", "blue"))
oplot_ev2.2
```



#EFS by ctDNA post-NAT - all stages

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Keio Neo FLOT_Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.postNAC!="",]
circ_datadf <- as.data.frame(circ_data)

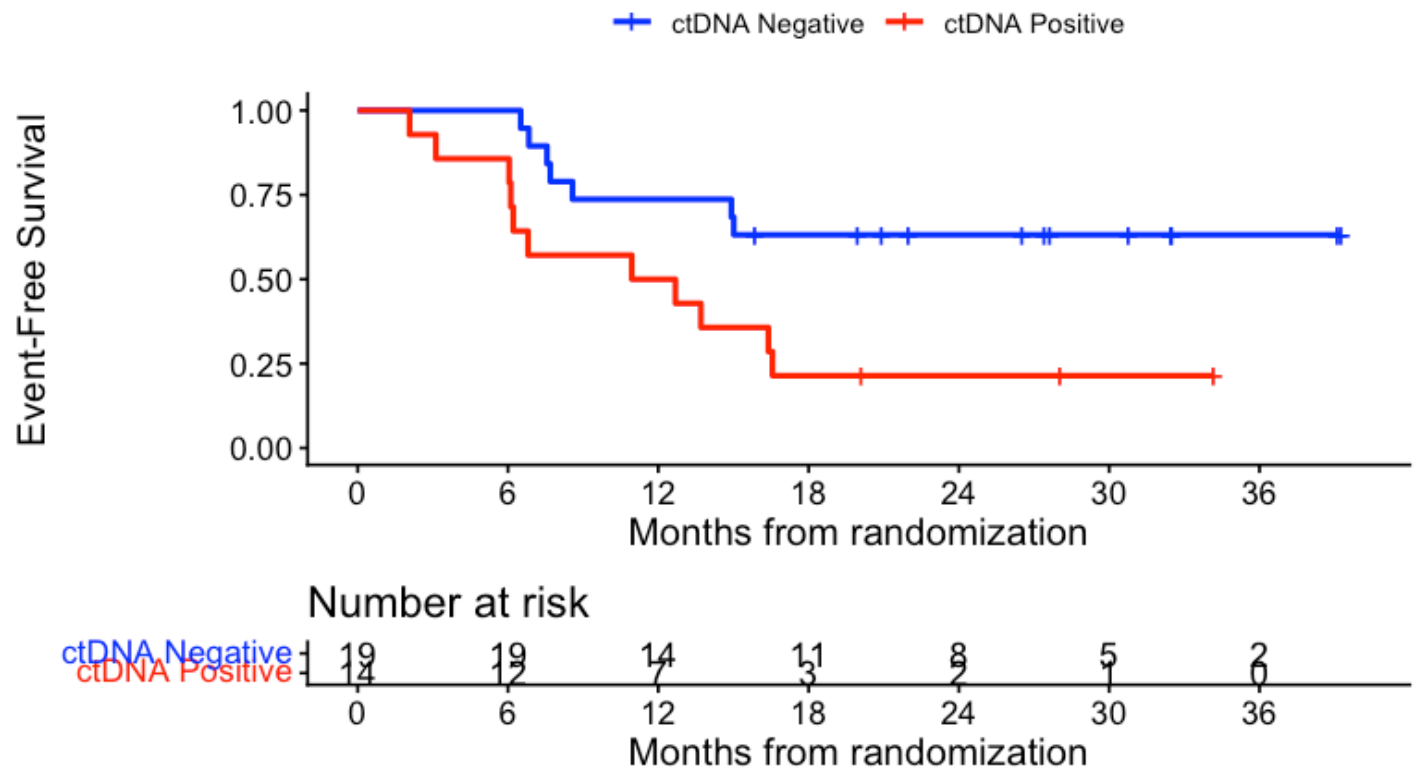
survfit(Surv(time = circ_data$EFS.months, event = circ_data$EFS.Event)~ctDNA.postNAC, data = circ_data)
```

Call: survfit(formula = Surv(time = circ\_data\$EFS.months, event = circ\_data\$EFS.Event) ~ ctDNA.postNAC, data = circ\_data)

|                        | n  | events | median | 0.95LCL | 0.95UCL |
|------------------------|----|--------|--------|---------|---------|
| ctDNA.postNAC=NEGATIVE | 19 | 7      | NA     | 15.01   | NA      |
| ctDNA.postNAC=POSITIVE | 14 | 11     | 11.8   | 6.21    | NA      |

```
surv_object <-Surv(time = circ_data$EFS.months, event = circ_data$EFS.Event)
KM_curve <- survfit(surv_object ~ ctDNA.postNAC, data = circ_data, conf.int=0.95, conf.type="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE, break.time.by=6, palette=c("blue","red"), title="EFS - ctDNA status post-NAT | All pts", ylab= "Event-Free Survival", xlab="Months from randomization", legend.labs=c("ctDNA Negative", "ctDNA Positive"), legend.title="")
```

EFS - ctDNA status post-NAT | All pts



Hide

```
summary(KM_curve, times= c(24))
```

Call: `survfit(formula = surv_object ~ ctDNA.postNAC, data = circ_data, conf.int = 0.95, conf.type = "log-log")`

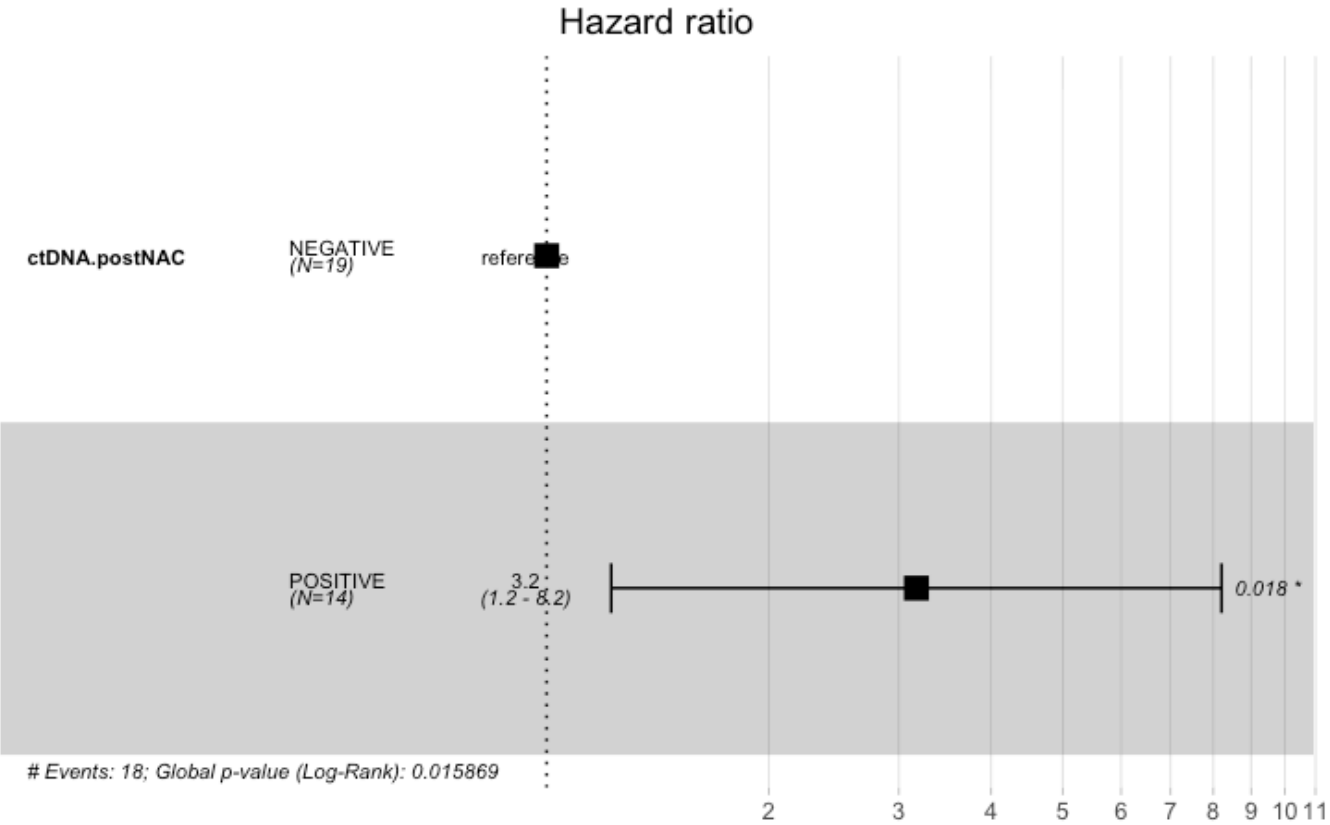
| ctDNA.postNAC=NEGATIVE |        |        |         |          |         |              |           |
|------------------------|--------|--------|---------|----------|---------|--------------|-----------|
|                        | time   | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% |
| CI                     |        |        |         |          |         |              |           |
| 04                     | 24.000 | 8.000  | 7.000   | 0.632    | 0.111   | 0.379        | 0.8       |

| ctDNA.postNAC=POSITIVE |         |        |         |          |         |              |           |
|------------------------|---------|--------|---------|----------|---------|--------------|-----------|
|                        | time    | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% |
| CI                     |         |        |         |          |         |              |           |
| 79                     | 24.0000 | 2.0000 | 11.0000 | 0.2143   | 0.1097  | 0.0521       | 0.44      |

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```
circ_data$ctDNA.postNAC <- factor(circ_data$ctDNA.postNAC, levels=c("NEGATIVE","POSITIVE"))
cox_fit <- coxph(surv_object ~ ctDNA.postNAC, data=circ_data)
ggforest(cox_fit,data = circ_data)
```



Hide

```
summary(cox_fit)
```

Call:  
coxph(formula = surv\_object ~ ctDNA.postNAC, data = circ\_data)

n= 33, number of events= 18

|                       | coef   | exp(coef) | se(coef) | z     | Pr(> z ) |
|-----------------------|--------|-----------|----------|-------|----------|
| ctDNA.postNACPOSITIVE | 1.1532 | 3.1684    | 0.4857   | 2.375 | 0.0176 * |

---  
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

|                       | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|-----------------------|-----------|------------|-----------|-----------|
| ctDNA.postNACPOSITIVE | 3.168     | 0.3156     | 1.223     | 8.208     |

Concordance= 0.65 (se = 0.055 )  
Likelihood ratio test= 5.82 on 1 df, p=0.02  
Wald test = 5.64 on 1 df, p=0.02  
Score (logrank) test = 6.27 on 1 df, p=0.01

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```
cox_fit_summary <- summary(cox_fit)

# Extract values for HR, 95% CI, and p-value
HR <- cox_fit_summary$coefficients[2]
lower_CI <- cox_fit_summary$conf.int[3]
upper_CI <- cox_fit_summary$conf.int[4]
p_value <- cox_fit_summary$coefficients[5]
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_CI, 2), "); p = ", round(p_value, 3))
print(label_text)
```

```
[1] "HR = 3.17 (1.22-8.21); p = 0.018"
```

#EFS by ctDNA Clearance post-NAT

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```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Keio Neo FLOT_Clinical Data.csv")

circ_data$ctDNA.C2D1.Clearance <- NA #first we create the variable for the ctDNA & NAC combination, and we assign values
circ_data <- circ_data %>%
  mutate(ctDNA.C2D1.Clearance = case_when(
    ctDNA.Base == "POSITIVE" & ctDNA.postNAC == "NEGATIVE" ~ "TRUE",
    ctDNA.Base == "POSITIVE" & ctDNA.postNAC == "POSITIVE" ~ "FALSE",
  ))

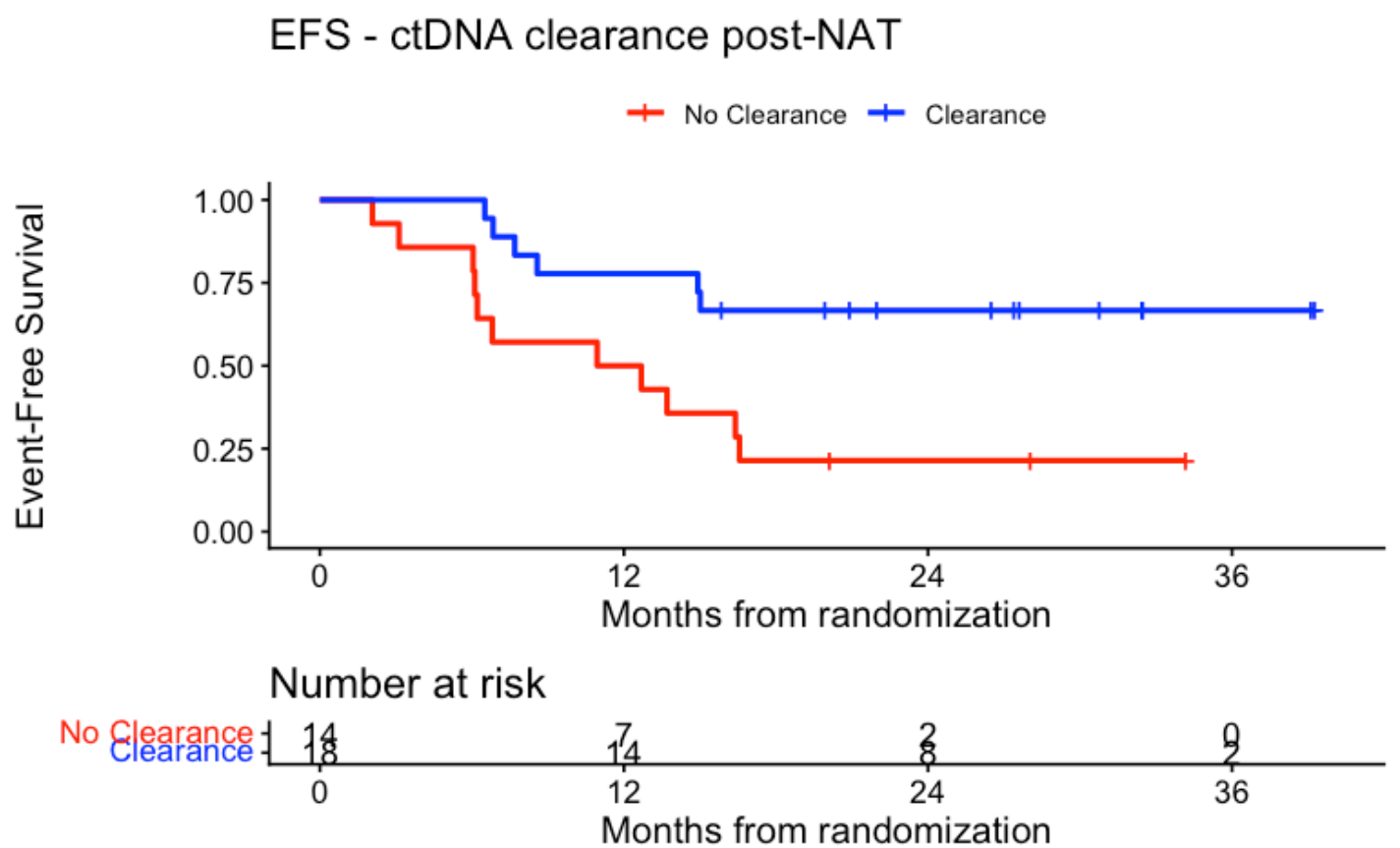
circ_data <- circ_data[!is.na(circ_data$ctDNA.C2D1.Clearance),]
survfit(Surv(time = circ_data$EFS.months, event = circ_data$EFS.Event)~ctDNA.C2D1.Clearance, data = circ_data)
```

Call: survfit(formula = Surv(time = circ\_data\$EFS.months, event = circ\_data\$EFS.Event) ~ ctDNA.C2D1.Clearance, data = circ\_data)

|                            | n  | events | median | 0.95LCL | 0.95UCL |
|----------------------------|----|--------|--------|---------|---------|
| ctDNA.C2D1.Clearance=FALSE | 14 | 11     | 11.8   | 6.21    | NA      |
| ctDNA.C2D1.Clearance=TRUE  | 18 | 6      | NA     | 15.01   | NA      |

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```
surv_object <- Surv(time = circ_data$EFS.months, event = circ_data$EFS.Event)
KM_curve <- survfit(surv_object ~ ctDNA.C2D1.Clearance, data = circ_data, conf.int=0.95, conf.type="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE, break.time.by=12, palette=c("red","blue"), title="EFS - ctDNA clearance post-NAT", ylab="Event-Free Survival", xlab="Months from randomization", legend.labs=c("No Clearance", "Clearance"), legend.title="")
```



Hide

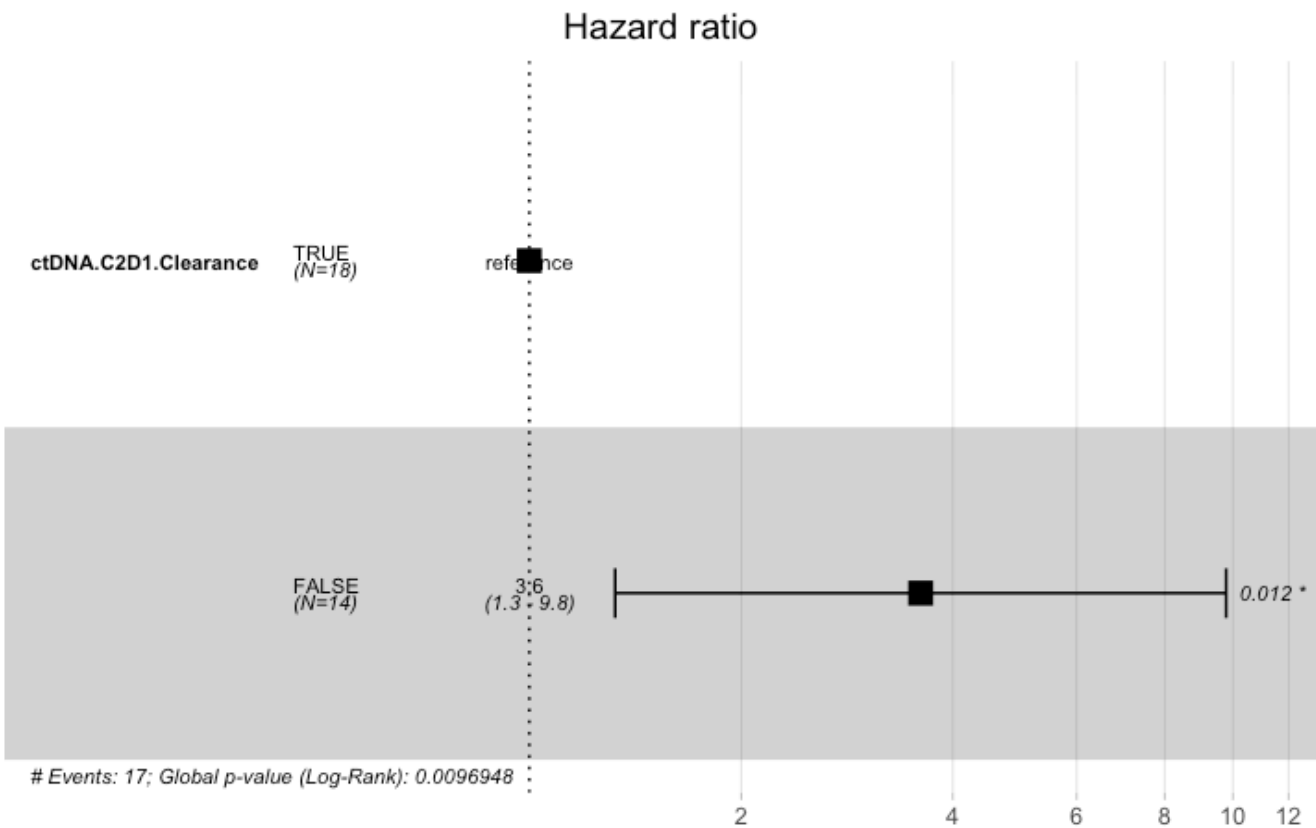
```
summary(KM_curve, times= c(24))
```

```
Call: survfit(formula = surv_object ~ ctDNA.C2D1.Clearance, data = circ_data,
  conf.int = 0.95, conf.type = "log-log")
```

| ctDNA.C2D1.Clearance=FALSE |         |        |         |          |         |              |           |
|----------------------------|---------|--------|---------|----------|---------|--------------|-----------|
|                            | time    | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% |
| CI                         |         |        |         |          |         |              |           |
| 79                         | 24.0000 | 2.0000 | 11.0000 | 0.2143   | 0.1097  | 0.0521       | 0.44      |
| ctDNA.C2D1.Clearance=TRUE  |         |        |         |          |         |              |           |
|                            | time    | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% |
| CI                         |         |        |         |          |         |              |           |
| 34                         | 24.000  | 8.000  | 6.000   | 0.667    | 0.111   | 0.404        | 0.8       |

Hide

```
circ_data$ctDNA.C2D1.Clearance <- factor(circ_data$ctDNA.C2D1.Clearance, levels=c("TRUE", "FALSE"))
cox_fit <- coxph(surv_object ~ ctDNA.C2D1.Clearance, data=circ_data)
ggforest(cox_fit, data = circ_data)
```



Hide

```
summary(cox_fit)
```

Call:

```
coxph(formula = surv_object ~ ctDNA.C2D1.Clearance, data = circ_data)
```

```
n= 32, number of events= 17
```

|                           | coef   | exp(coef) | se(coef) | z    | Pr(> z ) |
|---------------------------|--------|-----------|----------|------|----------|
| ctDNA.C2D1.ClearanceFALSE | 1.2809 | 3.5997    | 0.5102   | 2.51 | 0.0121 * |

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

|                           | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|---------------------------|-----------|------------|-----------|-----------|
| ctDNA.C2D1.ClearanceFALSE | 3.6       | 0.2778     | 1.324     | 9.785     |

Concordance= 0.665 (se = 0.055 )

Likelihood ratio test= 6.69 on 1 df, p=0.01

Wald test = 6.3 on 1 df, p=0.01

Score (logrank) test = 7.18 on 1 df, p=0.007

[Hide](#)

```
cox_fit_summary <- summary(cox_fit)
```

```
# Extract values for HR, 95% CI, and p-value
```

```
HR <- cox_fit_summary$coefficients[2]
```

```
lower_CI <- cox_fit_summary$conf.int[3]
```

```
upper_CI <- cox_fit_summary$conf.int[4]
```

```
p_value <- cox_fit_summary$coefficients[5]
```

```
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_C  
I, 2), "); p = ", round(p_value, 3))
```

```
print(label_text)
```

```
[1] "HR = 3.6 (1.32-9.78); p = 0.012"
```

#Association of ctDNA status with Pathological N status

[Hide](#)

```
rm(list=ls())
```

```
setwd("~/Downloads")
```

```
circ_data <- read.csv("Keio Neo FLOT_Clinical Data.csv")
```

```
circ_data$ctDNA.postNAC <- factor(circ_data$ctDNA.postNAC, levels = c("NEGATIVE", "POSIT  
IVE"), labels = c("Negative", "Positive"))
```

```
circ_data$pN.stage <- factor(circ_data$pN.stage, levels = c("N0", "N1-N3"), labels = c  
("N0", "N1-N3"))
```

```
contingency_table <- table(circ_data$pN.stage, circ_data$ctDNA.postNAC)
```

```
chi_square_test <- chisq.test(contingency_table)
```

```
print(chi_square_test)
```

### Pearson's Chi-squared test with Yates' continuity correction

```
data: contingency_table
X-squared = 3.4977, df = 1, p-value = 0.06145
```

[Hide](#)

```
fisher_exact_test <- fisher.test(contingency_table)
print(fisher_exact_test)
```

### Fisher's Exact Test for Count Data

```
data: contingency_table
p-value = 0.04638
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 0.9734627 471.1789080
sample estimates:
odds ratio
 9.250939
```

[Hide](#)

```
print(contingency_table)
```

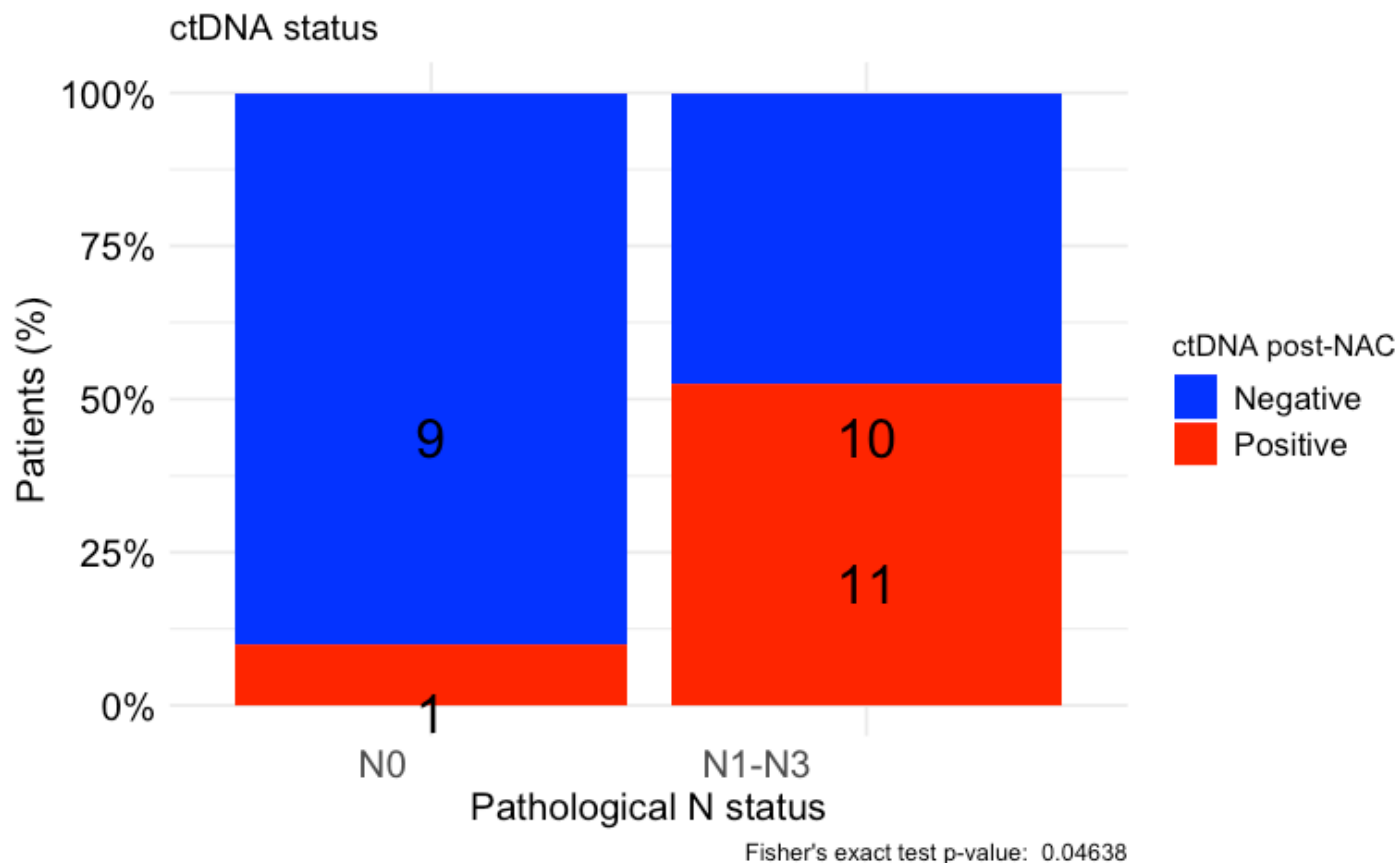
|       | Negative | Positive |
|-------|----------|----------|
| N0    | 9        | 1        |
| N1-N3 | 10       | 11       |

[Hide](#)

```

table_df <- as.data.frame(contingency_table)
table_df$Total <- ave(table_df$Freq, table_df$Var1, FUN = sum)
table_df$Percentage <- table_df$Freq / table_df$Total
table_df$MiddlePercentage <- table_df$Percentage / 2
ggplot(table_df, aes(x = Var1, y = Percentage, fill = Var2)) +
  geom_bar(stat = "identity") +
  geom_text(aes(y = MiddlePercentage, label = Freq), position = "stack", color = "black", vjust = 1.5, size = 7) +
  theme_minimal() +
  labs(title = "ctDNA status",
        x = "Pathological N status",
        y = "Patients (%)",
        fill = "ctDNA post-NAC",
        caption = paste("Fisher's exact test p-value: ", format.pval(fisher_exact_test$p.value))) +
  scale_y_continuous(labels = scales::percent_format()) +
  scale_fill_manual(values = c("Negative" = "blue", "Positive" = "red")) + # define custom colors
  theme(axis.text.x = element_text(angle = 0, hjust = 1.5, size = 14), # increase x-axis text size
        axis.text.y = element_text(size = 14, color = "black"), # increase y-axis text size
        axis.title.x = element_text(size = 14, color = "black"), # increase x-axis label size
        axis.title.y = element_text(size = 14, color = "black"), # increase y-axis label size
        legend.text = element_text(size = 12, color = "black")) # increase Recurrence label size

```



#Association of ctDNA clearance with Pathological N status

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Keio Neo FLOT_Clinical Data.csv")

circ_data$ctDNA.clearance <- NA
circ_data <- circ_data %>%
  mutate(ctDNA.clearance = case_when(
    ctDNA.Base == "POSITIVE" & ctDNA.postNAC == "NEGATIVE" ~ "TRUE",
    ctDNA.Base == "POSITIVE" & ctDNA.postNAC == "POSITIVE" ~ "FALSE",
  ))

circ_data <- circ_data[!is.na(circ_data$ctDNA.clearance),]

circ_data$ctDNA.clearance <- factor(circ_data$ctDNA.clearance, levels = c("TRUE", "FALSE"), labels = c("Clearance", "No Clearance"))
circ_data$pN.stage <- factor(circ_data$pN.stage, levels = c("N0", "N1-N3"), labels = c("N0", "N1-N3"))
contingency_table <- table(circ_data$pN.stage, circ_data$ctDNA.clearance)
chi_square_test <- chisq.test(contingency_table)
print(chi_square_test)
```

### Pearson's Chi-squared test with Yates' continuity correction

```
data: contingency_table
X-squared = 2.9167, df = 1, p-value = 0.08767
```

[Hide](#)

```
fisher_exact_test <- fisher.test(contingency_table)
print(fisher_exact_test)
```

### Fisher's Exact Test for Count Data

```
data: contingency_table
p-value = 0.04921
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 0.8452809 424.1258917
sample estimates:
odds ratio
 8.235925
```

[Hide](#)

```
print(contingency_table)
```

|       | Clearance | No Clearance |
|-------|-----------|--------------|
| N0    | 8         | 1            |
| N1-N3 | 10        | 11           |

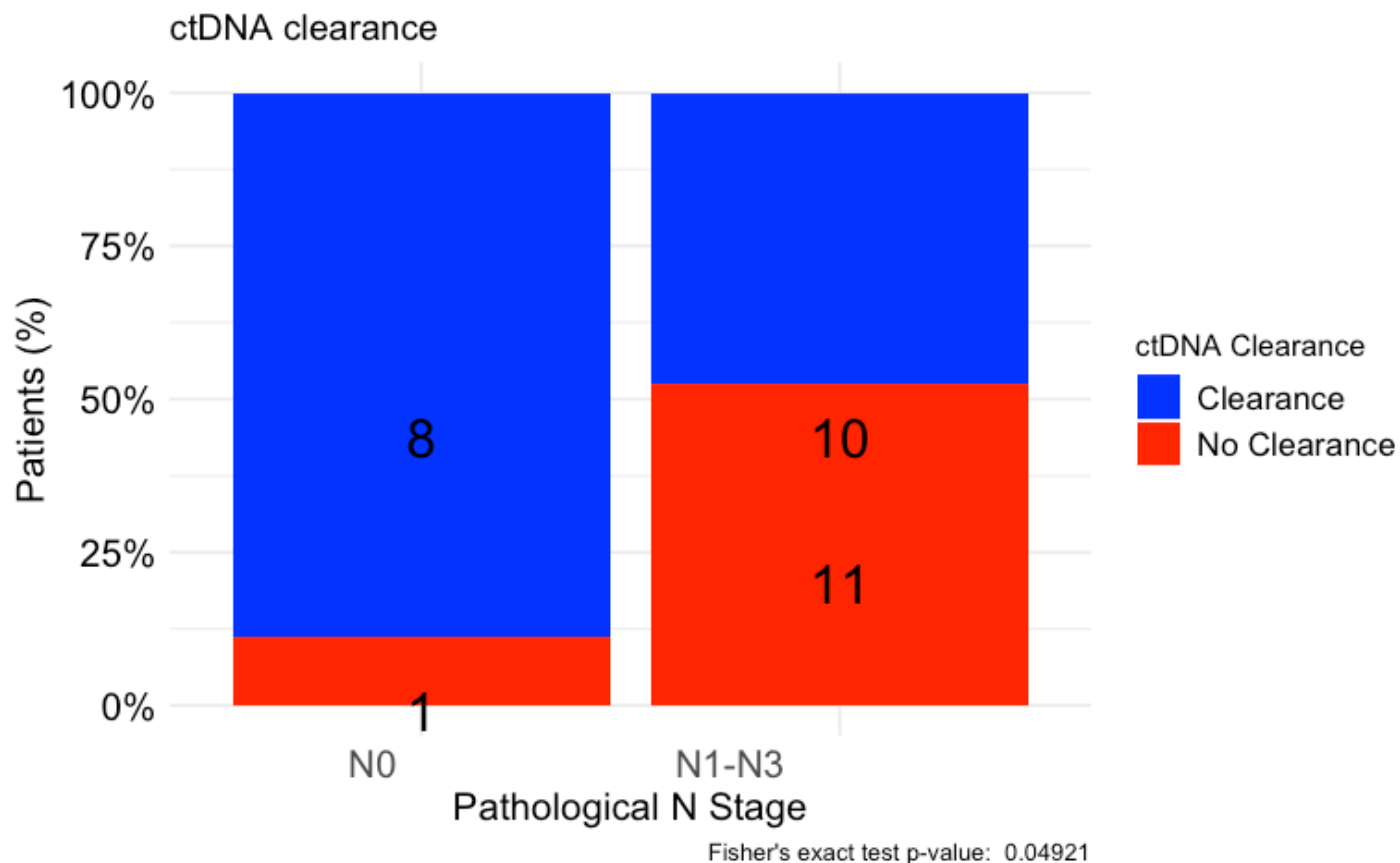
[Hide](#)



```

table_df <- as.data.frame(contingency_table)
table_df$Total <- ave(table_df$Freq, table_df$Var1, FUN = sum)
table_df$Percentage <- table_df$Freq / table_df$Total
table_df$MiddlePercentage <- table_df$Percentage / 2
ggplot(table_df, aes(x = Var1, y = Percentage, fill = Var2)) +
  geom_bar(stat = "identity") +
  geom_text(aes(y = MiddlePercentage, label = Freq), position = "stack", color = "black", vjust = 1.5, size = 7) +
  theme_minimal() +
  labs(title = "ctDNA clearance",
        x = "Pathological N Stage",
        y = "Patients (%)",
        fill = "ctDNA Clearance",
        caption = paste("Fisher's exact test p-value: ", format.pval(fisher_exact_test$p.value))) +
  scale_y_continuous(labels = scales::percent_format()) +
  scale_fill_manual(values = c("Clearance" = "blue", "No Clearance" = "red")) + # define custom colors
  theme(axis.text.x = element_text(angle = 0, hjust = 1.5, size = 14), # increase x-axis text size
        axis.text.y = element_text(size = 14, color = "black"), # increase y-axis text size
        axis.title.x = element_text(size = 14, color = "black"), # increase x-axis label size
        axis.title.y = element_text(size = 14, color = "black"), # increase y-axis label size
        legend.text = element_text(size = 12, color = "black")) # increase Recurrence label size

```



#Association of ctDNA dynamics with Pathological N status

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Keio Neo FLOT_Clinical Data.csv")

circ_data$ctDNA.Dynamics <- NA #first we create the variable for the ctDNA & NAC combina
tion, and we assign values
circ_data <- circ_data %>%
  mutate(ctDNA.Dynamics = case_when(
    ctDNA.Base == "POSITIVE" & ctDNA.postNAC == "NEGATIVE" & ctDNA.MRD == "NEGATIVE" ~
    "ctDNA +/-/-",
    ctDNA.Base == "POSITIVE" & ctDNA.postNAC == "POSITIVE" & ctDNA.MRD == "NEGATIVE" ~
    "ctDNA +/+/-",
    ctDNA.Base == "POSITIVE" & ctDNA.postNAC == "POSITIVE" & ctDNA.MRD == "POSITIVE" ~
    "ctDNA +/+/"
  ))

circ_data$ctDNA.Dynamics <- factor(circ_data$ctDNA.Dynamics, levels=c("ctDNA +/-/-", "ct
DNA +/+/-", "ctDNA +/+/" ))
circ_data$pN.stage <- factor(circ_data$pN.stage, levels = c("N0", "N1-N3"), labels = c
("N0", "N1-N3"))
contingency_table <- table(circ_data$ctDNA.Dynamics, circ_data$pN.stage)
chi_square_test <- chisq.test(contingency_table)
print(chi_square_test)
```

## Pearson's Chi-squared test

```
data: contingency_table
X-squared = 7.8431, df = 2, p-value = 0.01981
```

Hide

```
fisher_exact_test <- fisher.test(contingency_table)
print(fisher_exact_test)
```

## Fisher's Exact Test for Count Data

```
data: contingency_table
p-value = 0.01785
alternative hypothesis: two.sided
```

Hide

```
print(contingency_table)
```

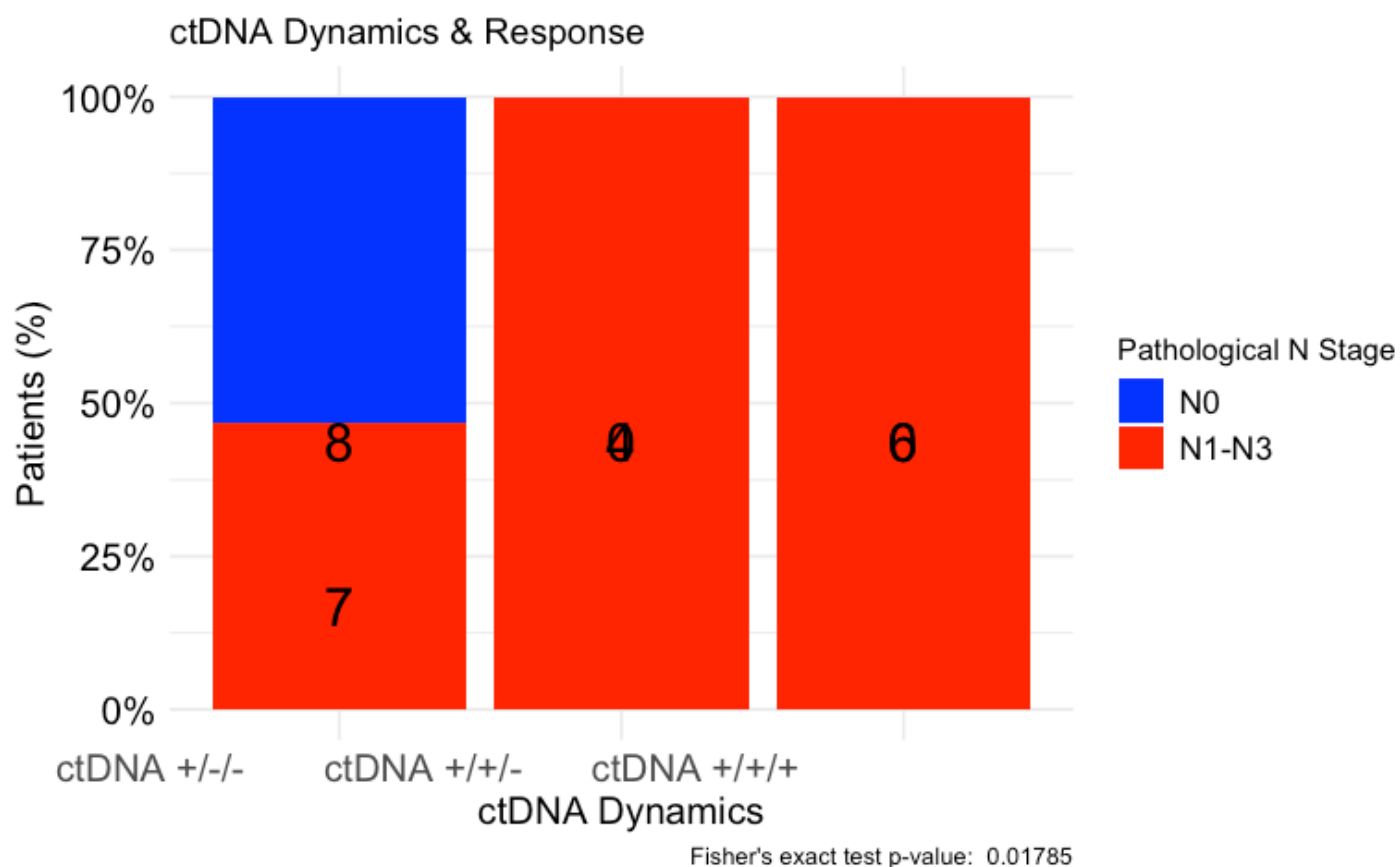
|             | N0 | N1-N3 |
|-------------|----|-------|
| ctDNA +/-/- | 8  | 7     |
| ctDNA +/+/- | 0  | 4     |
| ctDNA +/+/+ | 0  | 6     |

Hide

```

table_df <- as.data.frame(contingency_table)
table_df$Total <- ave(table_df$Freq, table_df$Var1, FUN = sum)
table_df$Percentage <- table_df$Freq / table_df$Total
table_df$MiddlePercentage <- table_df$Percentage / 2
ggplot(table_df, aes(x = Var1, y = Percentage, fill = Var2)) +
  geom_bar(stat = "identity") +
  geom_text(aes(y = MiddlePercentage, label = Freq), position = "stack", color = "black",
    vjust = 1.5, size = 7) +
  theme_minimal() +
  labs(title = "ctDNA Dynamics & Response",
    x = "ctDNA Dynamics",
    y = "Patients (%)",
    fill = "Pathological N Stage",
    caption = paste("Fisher's exact test p-value: ", format.pval(fisher_exact_test$p.value))) +
  scale_y_continuous(labels = scales::percent_format()) +
  scale_fill_manual(values = c("N0" = "blue", "N1-N3" = "red")) + # define custom colors
  theme(axis.text.x = element_text(angle = 0, hjust = 1.5, size = 14), # increase x-axis
    text size
    axis.text.y = element_text(size = 14, color = "black"), # increase y-axis text size
    axis.title.x = element_text(size = 14, color = "black"), # increase x-axis label size
    axis.title.y = element_text(size = 14, color = "black"), # increase y-axis label size
    legend.text = element_text(size = 12, color = "black")) # increase Recurrence label size

```



## #EFS by ctDNA at the MRD Window

Hide

```
rm(list=ls())
setwd("~/Downloads")
circ_data <- read.csv("Keio Neo FLOT_Clinical Data.csv")
circ_data <- circ_data[circ_data$ctDNA.MRD!="",]

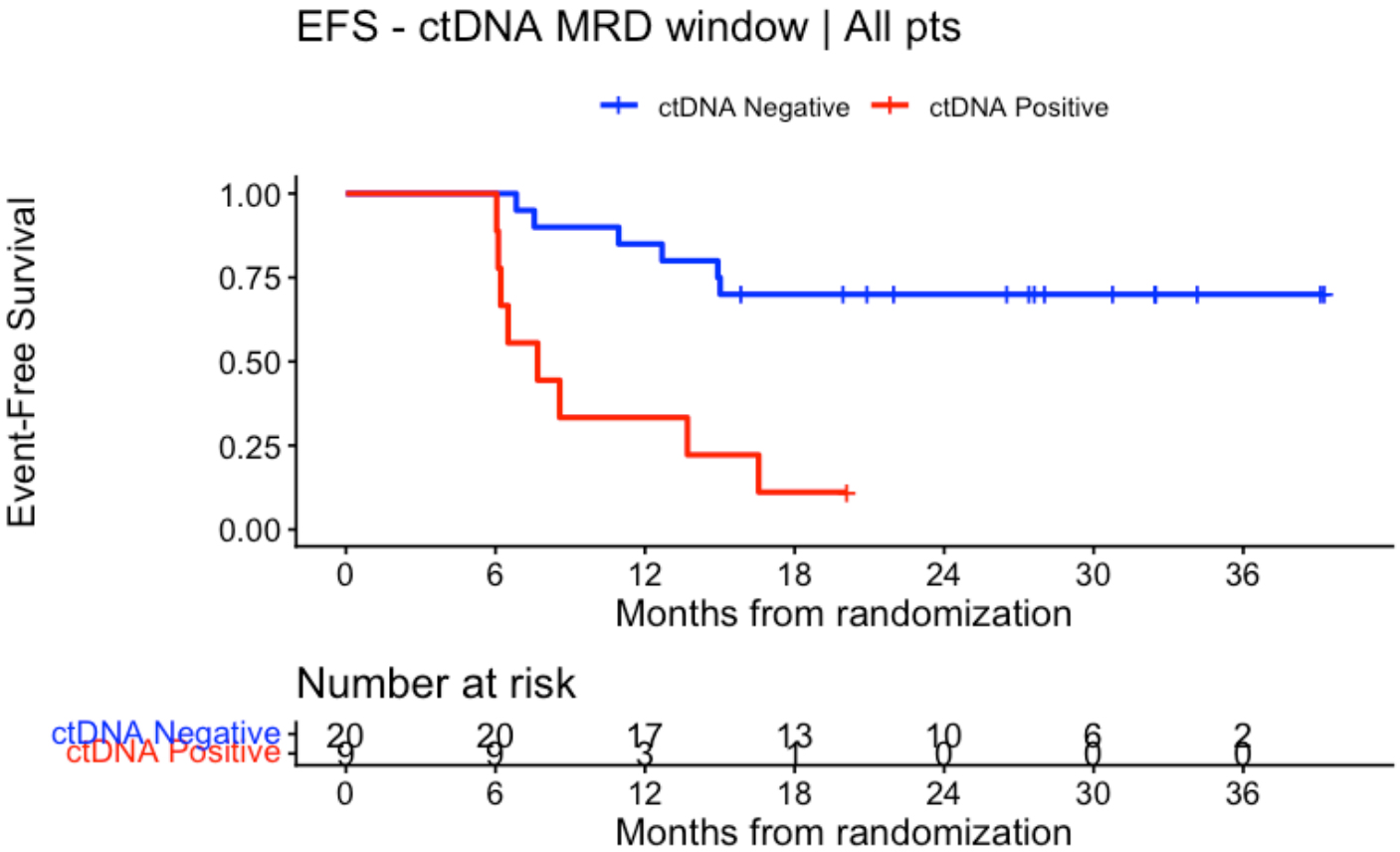
survfit(Surv(time = circ_data$EFS.months, event = circ_data$EFS.Event)~ctDNA.MRD, data =
circ_data)
```

Call: survfit(formula = Surv(time = circ\_data\$EFS.months, event = circ\_data\$EFS.Event) ~  
ctDNA.MRD, data = circ\_data)

|                    | n  | events | median | 0.95LCL | 0.95UCL |
|--------------------|----|--------|--------|---------|---------|
| ctDNA.MRD=NEGATIVE | 20 | 6      | NA     | NA      | NA      |
| ctDNA.MRD=POSITIVE | 9  | 8      | 7.69   | 6.21    | NA      |

Hide

```
surv_object <-Surv(time = circ_data$EFS.months, event = circ_data$EFS.Event)
KM_curve <- survfit(surv_object ~ ctDNA.MRD, data = circ_data,conf.int=0.95,conf.type="l
og-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRU
E, break.time.by=6, palette=c("blue","red"), title="EFS - ctDNA MRD window | All pts", y
lab= "Event-Free Survival", xlab="Months from randomization", legend.labs=c("ctDNA Negat
ive", "ctDNA Positive"), legend.title="")
```



Hide

```
summary(KM_curve, times= c(0, 24))
```

```
Call: survfit(formula = surv_object ~ ctDNA.MRD, data = circ_data,
  conf.int = 0.95, conf.type = "log-log")
```

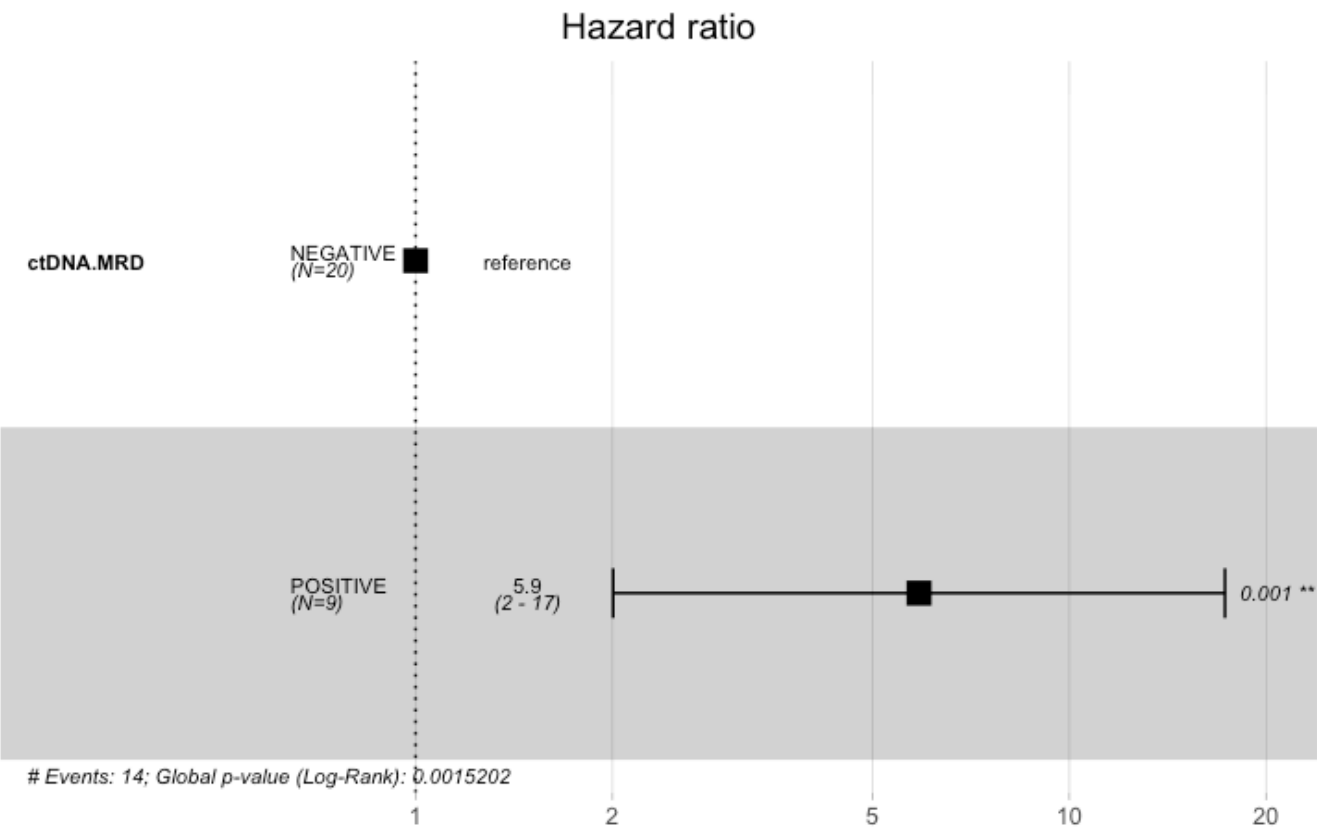
| ctDNA.MRD=NEGATIVE |        |         |          |         |              |              |  |
|--------------------|--------|---------|----------|---------|--------------|--------------|--|
| time               | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |  |
| 0                  | 20     | 0       | 1.0      | 0.000   | 1.000        | 1.000        |  |
| 24                 | 10     | 6       | 0.7      | 0.102   | 0.451        | 0.853        |  |

| ctDNA.MRD=POSITIVE |      |        |         |          |         |              |              |
|--------------------|------|--------|---------|----------|---------|--------------|--------------|
| CI                 | time | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |
| 1                  | 0    | 9      | 0       | 1        | 0       | 1            |              |

Hide

```
circ_data$ctDNA.MRD <- factor(circ_data$ctDNA.MRD, levels=c("NEGATIVE","POSITIVE"))
cox_fit <- coxph(surv_object ~ ctDNA.MRD, data=circ_data)
ggforest(cox_fit,data = circ_data)
```



Hide

```
summary(cox_fit)
```

Call:

```
coxph(formula = surv_object ~ ctDNA.MRD, data = circ_data)
```

n= 29, number of events= 14

|                   | coef   | exp(coef) | se(coef) | z     | Pr(> z )   |
|-------------------|--------|-----------|----------|-------|------------|
| ctDNA.MRDPOSITIVE | 1.7732 | 5.8898    | 0.5498   | 3.225 | 0.00126 ** |

----

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

|                   | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|-------------------|-----------|------------|-----------|-----------|
| ctDNA.MRDPOSITIVE | 5.89      | 0.1698     | 2.005     | 17.3      |

Concordance= 0.708 (se = 0.055 )

Likelihood ratio test= 10.05 on 1 df, p=0.002

Wald test = 10.4 on 1 df, p=0.001

Score (logrank) test = 13.14 on 1 df, p=3e-04

[Hide](#)

```
cox_fit_summary <- summary(cox_fit)
```

```
# Extract values for HR, 95% CI, and p-value
```

```
HR <- cox_fit_summary$coefficients[2]
```

```
lower_CI <- cox_fit_summary$conf.int[3]
```

```
upper_CI <- cox_fit_summary$conf.int[4]
```

```
p_value <- cox_fit_summary$coefficients[5]
```

```
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_CI, 2), "); p = ", round(p_value, 3))
```

```
print(label_text)
```

```
[1] "HR = 5.89 (2.01-17.3); p = 0.001"
```

#OS by ctDNA at the MRD Window

[Hide](#)

```
rm(list=ls())
```

```
setwd("~/Downloads")
```

```
circ_data <- read.csv("Keio Neo FLOT_Clinical Data.csv")
```

```
circ_data <- circ_data[circ_data$ctDNA.MRD!="",]
```

```
survfit(Surv(time = circ_data$OS.months, event = circ_data$OS.Event)~ctDNA.MRD, data = circ_data)
```



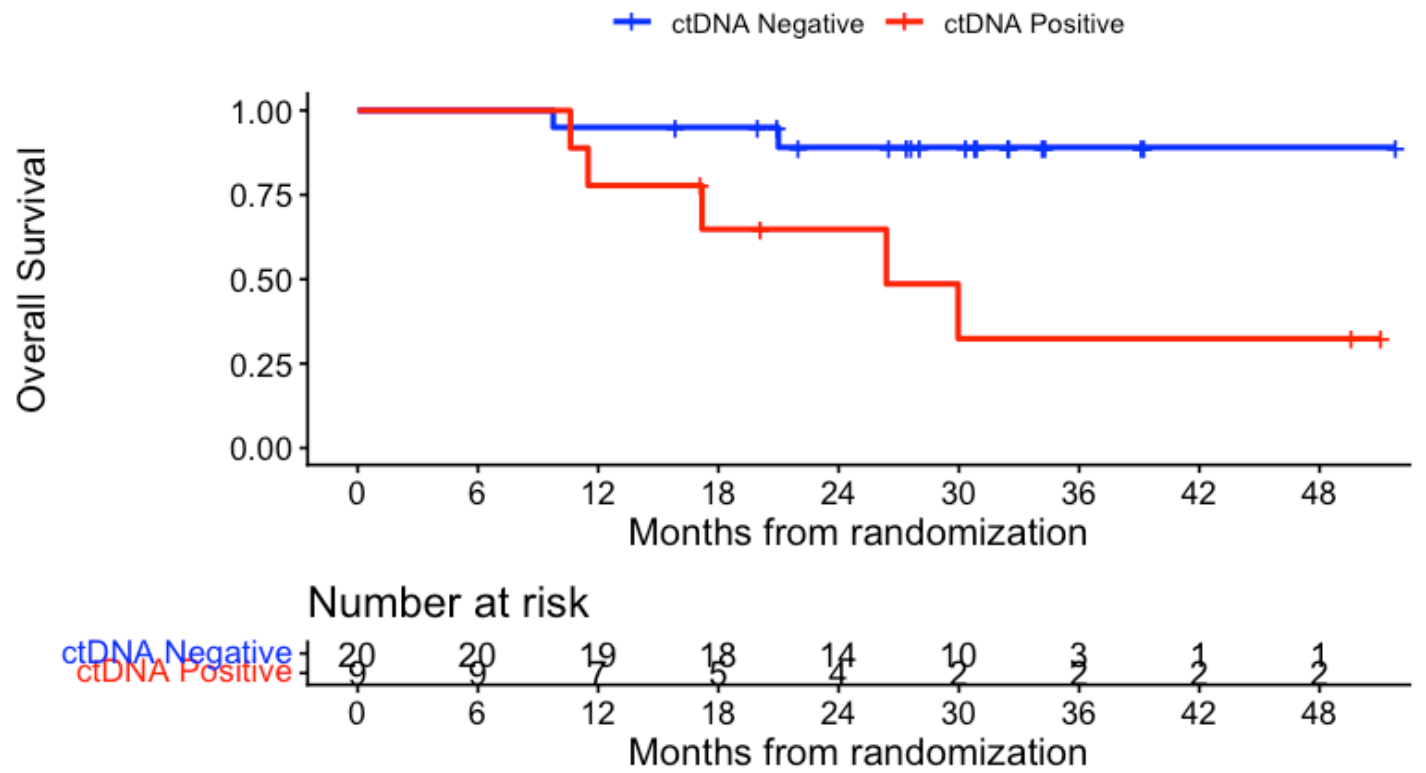
```
Call: survfit(formula = Surv(time = circ_data$OS.months, event = circ_data$OS.Event) ~
      ctDNA.MRD, data = circ_data)
```

|                    | n  | events | median | 0.95LCL | 0.95UCL |
|--------------------|----|--------|--------|---------|---------|
| ctDNA.MRD=NEGATIVE | 20 | 2      | NA     | NA      | NA      |
| ctDNA.MRD=POSITIVE | 9  | 5      | 26.4   | 17.2    | NA      |

[Hide](#)

```
surv_object <-Surv(time = circ_data$OS.months, event = circ_data$OS.Event)
KM_curve <- survfit(surv_object ~ ctDNA.MRD, data = circ_data,conf.int=0.95,conf.type="log-log")
ggsurvplot(KM_curve, data = circ_data, pval = FALSE, conf.int = FALSE, risk.table = TRUE, break.time.by=6, palette=c("blue","red"), title="OS - ctDNA MRD window | All pts", ylab= "Overall Survival", xlab="Months from randomization", legend.labs=c("ctDNA Negative", "ctDNA Positive"), legend.title="")
```

OS - ctDNA MRD window | All pts



Hide

```
summary(KM_curve, times= c(0, 24))
```

```
Call: survfit(formula = surv_object ~ ctDNA.MRD, data = circ_data,
  conf.int = 0.95, conf.type = "log-log")
```

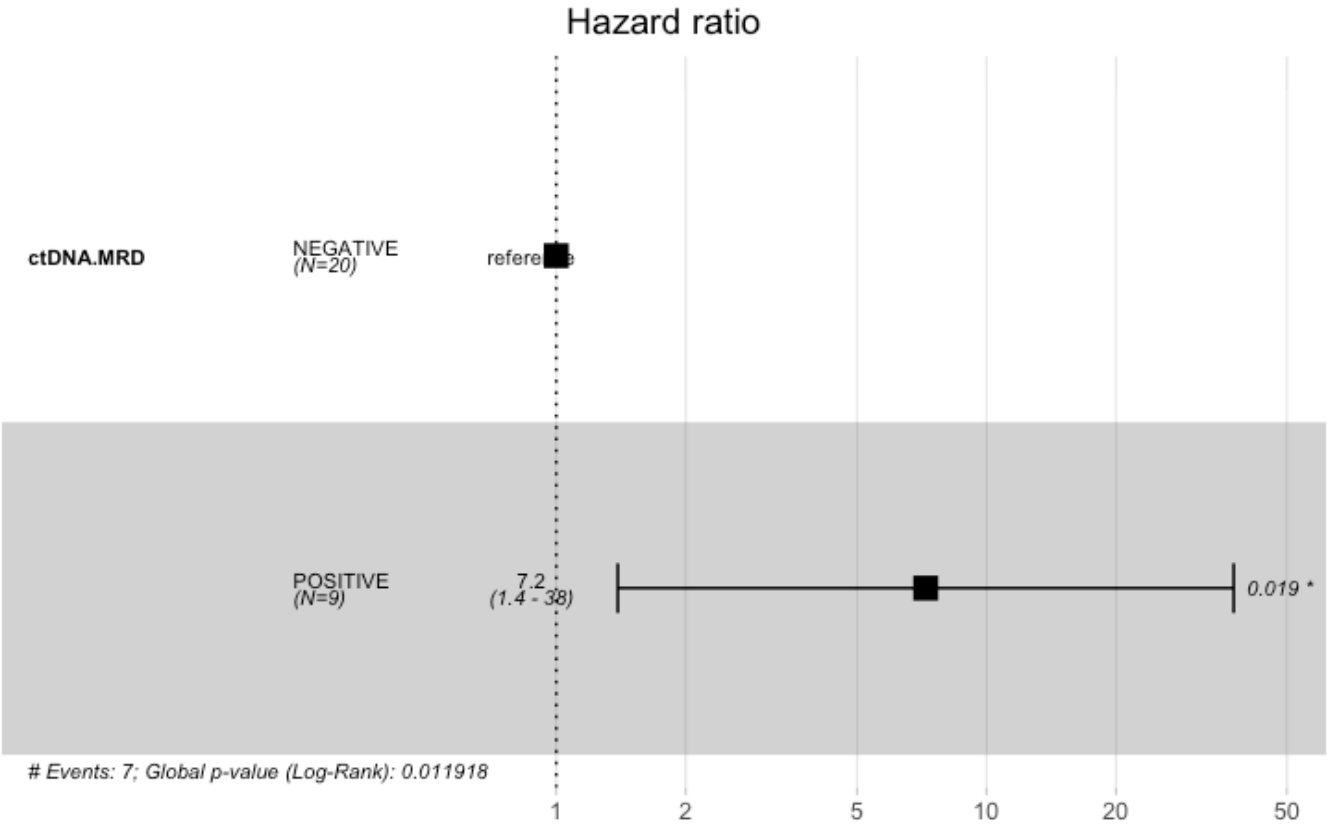
| ctDNA.MRD=NEGATIVE |        |         |          |         |              |              |  |
|--------------------|--------|---------|----------|---------|--------------|--------------|--|
| time               | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |  |
| 0                  | 20     | 0       | 1.000    | 0.0000  | 1.000        | 1.000        |  |
| 24                 | 14     | 2       | 0.891    | 0.0734  | 0.627        | 0.972        |  |

| ctDNA.MRD=POSITIVE |        |         |          |         |              |              |  |
|--------------------|--------|---------|----------|---------|--------------|--------------|--|
| time               | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |  |
| 0                  | 9      | 0       | 1.000    | 0.000   | 1.000        | 1.000        |  |
| 24                 | 4      | 3       | 0.648    | 0.165   | 0.253        | 0.872        |  |

Hide

```
circ_data$ctDNA.MRD <- factor(circ_data$ctDNA.MRD, levels=c("NEGATIVE","POSITIVE"))
cox_fit <- coxph(surv_object ~ ctDNA.MRD, data=circ_data)
ggforest(cox_fit,data = circ_data)
```



Hide

```
summary(cox_fit)
```

Call:  
coxph(formula = surv\_object ~ ctDNA.MRD, data = circ\_data)

n= 29, number of events= 7

|                   | coef   | exp(coef) | se(coef) | z     | Pr(> z ) |
|-------------------|--------|-----------|----------|-------|----------|
| ctDNA.MRDPOSITIVE | 1.9779 | 7.2277    | 0.8411   | 2.351 | 0.0187 * |

---  
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

|                   | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|-------------------|-----------|------------|-----------|-----------|
| ctDNA.MRDPOSITIVE | 7.228     | 0.1384     | 1.39      | 37.58     |

Concordance= 0.72 (se = 0.093 )  
Likelihood ratio test= 6.32 on 1 df, p=0.01  
Wald test = 5.53 on 1 df, p=0.02  
Score (logrank) test = 7.5 on 1 df, p=0.006

Hide

```
cox_fit_summary <- summary(cox_fit)

# Extract values for HR, 95% CI, and p-value
HR <- cox_fit_summary$coefficients[2]
lower_CI <- cox_fit_summary$conf.int[3]
upper_CI <- cox_fit_summary$conf.int[4]
p_value <- cox_fit_summary$coefficients[5]
label_text <- paste0("HR = ", round(HR, 2), " (", round(lower_CI, 2), "-", round(upper_C
I, 2), "); p = ", round(p_value, 3))
print(label_text)
```

```
[1] "HR = 7.23 (1.39-37.58); p = 0.019"
```